

PHYSICS

1. An ideal gas with an adiabatic exponent 1.5, initially at 27°C is compressed adiabatically from 800 cc to 200 cc. The final temperature of the gas is

- (1) 600 K
- (2) 300 K
- (3) 450 K
- (4) 273 K

Ans. (1)

Sol. $PV^2 = \text{constant}$

$$TV^{r-3} = \text{constant}$$

$$T_1 V_1^{r-1} = T_2 V_2^{r-1}$$

$$T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{r-1} = (300) \left(\frac{800}{200} \right)^{\frac{1}{2}}$$

$$= 600 \text{ K}$$

2. In YDSE, light of intensity of 4I and 9I passes through two slits respectively. Difference of maximum and minimum intensity of interference pattern is

- (1) 5I
- (2) 10I
- (3) 24I
- (4) 26I

Ans. (3)

Sol. $I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2 = (5\sqrt{I})^2 = 25I$

$$I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2 = (\sqrt{I})^2 = I$$

$$\Delta I = 24 I$$

3. Choose the correct option.

- | | |
|--------------------------------|-----------------------|
| a. Gravitational potential | (i) $M^{-1}L^3T^{-2}$ |
| b. Gravitational constant | (ii) ML^2T^{-2} |
| c. Acceleration due to gravity | (iii) $M^0L^2T^{-2}$ |
| d. Potential energy | (iv) M^0LT^{-2} |

- (1) a(iii), b(ii), c(iv), d(i)
- (2) a(iii), b(i), c(iv), d(ii)
- (3) a(ii), b(i), c(iv), d(iii)
- (4) a(ii), b(iv), c(i), d(iii)

Ans. (2)

Sol. Conceptual.

4. The work function of a metal 3eV. The colour of the visible light that is required to cause emission of photo electrons is

- (1) Yellow
- (2) Blue
- (3) Red
- (4) Green

Ans. (2)

Sol. $\phi = 3\text{ eV}$

$$\frac{hc}{\lambda} = 3\text{ eV} \Rightarrow \lambda = \frac{1240}{3\text{ eV}} \text{ eVnm}$$

$$= 413 \text{ nm}$$

$$\text{Blue} = 380\text{--}450 \text{ nm}$$

- 5.** Power of point source is 450 watt. Radiation pressure on a perfectly reflecting surface at a distance of 2 m is

(1) 1.5×10^{-8}

(2) 3×10^{-8}

(3) 0

(4) 6×10^{-8}

Ans. (4)

Sol. $I = I = \frac{P}{4\pi r^2} = \frac{450}{4\pi(2)^2}$

$$\text{Radiation Pressure} = \frac{2I}{C} = \frac{2 \times 450}{4\pi \times 4 \times 3 \times 10^8}$$

$$\frac{150}{8\pi} \times 10^{-8}$$

$$= 6 \times 10^{-8}$$

- 6.** A force of 49N acts tangentially at the highest point of a sphere (solid of mass 20 kg) kept on a rough horizontal plane. If the sphere rolls without slipping, then the acceleration of the center of the sphere is

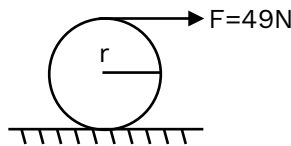
(1) 0.25 m/s^2

(2) 2.5 m/s^2

(3) 3.5 m/s^2

(4) 0.35 m/s^2

Ans. (3)



Sol. $49 - f = 20a$

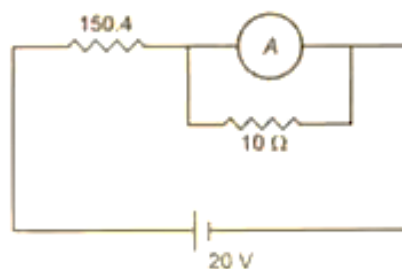
$$(49 + f)R = \frac{2}{5} \times 20 \times R^2 \frac{a}{R}$$

$$98 = 28a$$

$$a = \frac{98}{28}$$

$$a = 3.5 \text{ m/s}^2$$

- 7.** An ammeter having resistance 240Ω is connected in the given circuit as shown. Find current through the ammeter.



(1) 1 mA

(3) 100 mA

(2) 5 mA

(4) 2.5 mA

Ans. (2)

Sol. $R_{eq} = 150.4 + \frac{240 \times 10}{250} = 160 \Omega$

$$i = \frac{20}{160 \Omega}$$

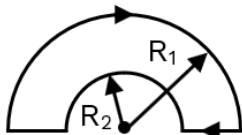
$$i = \frac{1}{8} \text{ A}$$

$$i_A = \frac{1}{8} \times \frac{10}{10 + 240} = \frac{1}{8 \times 25} = \frac{1}{200} = 5 \text{ mA}$$

- 8.** A thin uniform wire of length 25 m and area of cross-section 5 mm^2 has resistivity $2 \times 10^{-6} \Omega \text{ m}$. If the wire is bent to form a circle, the resistance across diametrically opposite points is

Sol. $\frac{\rho l}{A} = \frac{2 \times 10^{-6} \times 25}{5 \times 10^{-6} \times 4} = \frac{5}{2} \Omega$
 $= 2.5 \Omega$

- 9.** $I = 20 \text{ A}$



$$R_1 = 6\pi, R_2 = 4\pi$$

Find magnetic field at point 'O'.

Sol. $B_{net} = \frac{\mu_0 I}{4R_2} - \frac{\mu_0 I}{4R_1}$

$$\frac{\mu_0 I}{4} \left[\frac{1}{R_2} - \frac{1}{R_1} \right]$$

$$= \frac{\mu_0 \times 20}{4} \times \left[\frac{1}{4\pi} - \frac{1}{6\pi} \right]$$

$$\frac{\mu_0}{4\pi} 20 \left[\frac{1}{4} - \frac{1}{6} \right]$$

$$= 10^{-7} 20 \times \frac{2}{12}$$

$$= 3.33 \times 10^{-7} \text{ T}$$

- 10.** A particle is released from height 's' above the surface of the earth. At certain height its K.E is 3 times of P.E. The height from the surface of the earth and the speed of the Particle at the instant are respectively.

Sol. $K_1 = 0, P_1 = mgs$
 $K_2 = 3P_2 \Rightarrow P_2 = mgh$
 $K_1 + P_1 = K_2 + P_2$
 $mgs = 4 mgh$
 $h = s/4$

$$K_2 = \frac{1}{2} m V^2 = 3mg \frac{s}{4}$$

$$V = \sqrt{\frac{3gs}{2}}$$

- 11.** Biconvex lens having focal length 12 cm and the radius of curvature of two surfaces is given as 15 cm and 10 cm, find the R.I. of the lens.

Sol. $f = 12 \text{ cm}$

$$R_1 = 15 \text{ cm}$$

$$R_2 = 10 \text{ cm}$$

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{1}{12} = (\mu - 1) \left(\frac{1}{15} + \left(\frac{+1}{10} \right) \right)$$

$$\frac{1}{12} = (\mu - 1) \left(\frac{5}{30} \right)$$

$$\mu = \frac{3}{2}$$

- 12.** A particle is thrown at an angle ϕ from vertical from level. Draw the graph of $H_{\max}$ attained by the particle as a function of ϕ .

Sol. $H_{\max} = \frac{u_y^2}{2g} = \frac{V^2 \cos^2 \phi}{2g}$

$$V^2 \cos^2 \phi = 2gH_{\max}$$

$$H_{\max} \propto \cos^2 \phi$$

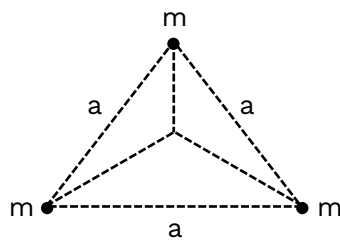
- 13.** The Electrostatic potential on the surface of uniformly charged spherical shell of radius $R = 10 \text{ cm}$ is 120 V. The potential at the centre of shell, at a distance 5 cm from centre and a distance 15 cm from the centre of the shells are

Sol. $V = \frac{KQ}{r} = \frac{KQ}{R}$

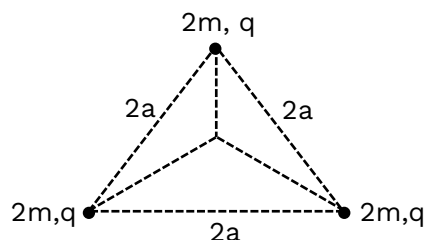
$$= 120 \times \frac{10}{5}$$

$$V_{15\text{cm}} = 80 \text{ volt}$$

14.



These mass particles collide because of gravitational attraction at $t = 4 \text{ sec.}$ if separation and mass becomes doubles then time taken for collision.



Sol. $m \rightarrow 2 \text{ times}$
 $r \rightarrow 2 \text{ times}$

$$F \rightarrow \text{Same}$$

$$F = mas$$

$$a = \frac{F}{2m} \Rightarrow \frac{a_0}{2}$$

$$S = ut + \frac{1}{2}at^2$$

$$S = \frac{1}{2}at^2$$

$$2a = \frac{1}{2} \frac{a_0}{2} t^2$$

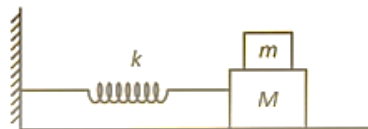
$$S \rightarrow 2s$$

$$a \rightarrow \frac{a_0}{2}$$

$$t^2 \rightarrow 4t_0^2$$

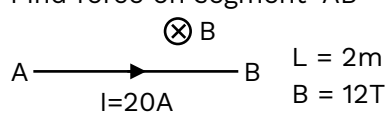
$$t \rightarrow 2t_0$$

- 15.** The figure below shows an oscillating system of two blocks and a spring. The horizontal surface is smooth and the contact between the blocks is rough with coefficient of static friction μ . Considering that the blocks of mass m is always stationary relative to M , choose the correct option regarding the statement below :



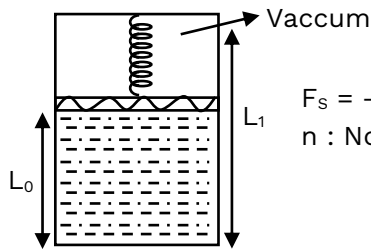
- (A) Maximum frictional force between blocks is μmg .
 (B) Time period of oscillation is $2\pi\sqrt{\frac{m+M}{K}}$
 (C) Friction between the blocks at any instant is $\mu(m+M)g$
 (1) Only A is correct
 (2) Only B is correct
 (3) A, B and C all three are correct

Sol. $f = ma$
 $f_{\max} = m\omega^2 A$
 $2\pi\sqrt{\frac{m+M}{K}}$

- 16.** Find force on segment 'AB'
- 

Sol. $B = 12T$
 $L = 2m$
 $I = 20A$
 $F = BIL$
 $= 12 \times 20 \times 2$
 $= 480N$

17.



$F_s = -kx^3$ (spring force)
 n : No. of moles of gas.

A gas is expanded isothermally from L_0 to L_1 . Find the amount of energy absorbed in given process.

Sol.

$$Q = W_{\text{gas}}$$

$$\Delta U = 0$$

$$\Delta T = 0$$

$$W_{\text{SP}} + W_g + W_{\text{gas}} = \Delta \text{KE}$$

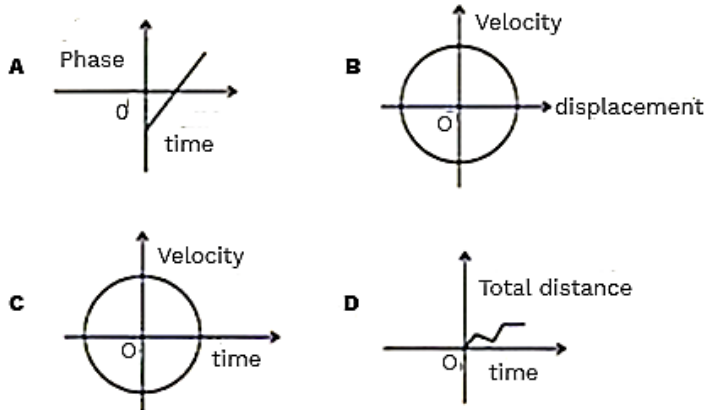
$$W_{\text{gas}} = -W_{\text{SP}} - W_g$$

$$W_{\text{SP}} = \int_{L_0}^{L_1} -Kx^3 dx$$

$$W_g = -mg(L_1 - L_0)$$

18.

Which of the following curved possibly represent one dimensional motion of a particle?



(1) A and B

(2) A,B,D

(3) A,B,C

(4) A,C,D

Sol.

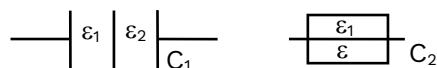
$$V^2 = \omega^2(A^2 - x^2)$$

$$V^2 = (A^2 - x^2)$$

$$V^2 + x^2 = A^2$$

19.

What is the ratio of C_1 and C_2 .



Sol.

$$C_1 = \frac{\frac{A\epsilon_1}{d/2} \times \frac{A\epsilon_2}{d/2}}{\frac{A\epsilon_1}{d/2} + \frac{A\epsilon_2}{d/2}}$$

$$= \frac{2A\epsilon_1\epsilon_2}{d(\epsilon_1 + \epsilon_2)}$$

$$C_2 = \frac{\frac{A}{2}\epsilon_1}{d} + \frac{\frac{A}{2}\epsilon_2}{d}$$

$$C_2 = \frac{A}{2d}(\epsilon_1 + \epsilon_2)$$

$$\frac{C_1}{C_2} = \frac{4\epsilon_1\epsilon_2}{(\epsilon_1 + \epsilon_2)^2}$$